Sing-Hao Ku

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Research Interests

Health Economics · Economic Demography · Social Policy

Education

New York University (NYU) New York, NY
M.A. in Economics; GPA: 3.81/4.00 09/2024 – 05/2026

- Concentration: Econometrics and Quantitative Economics
- Relevant Coursework: Math for Economists, Microeconomics, Macroeconomics, Applied Statistics and Econometrics, Machine Learning in Economics, Rationality and Behavioral Economics, Empirical Research Methods, Messy Data and Machine Learning

National Taiwan University (NTU) Taipei, Taiwan
B.B.A. in International Business, Minor in Economics; GPA: 3.75/4.30 (3.65/4.00) 09/2019 – 06/2023

Master's Thesis

Coverage Substitution After Job Loss: How the ACA Altered Post-Displacement Insurance Trajectories
M.A. Thesis · Advisor: Prof. Gustavo Pereira · New York University · Highest Grade in Cohort 01/2026 – 05/2026
[Full paper]

- Examines whether the Affordable Care Act (ACA), through its Medicaid expansion and Marketplace coverage, provides more protection for insurance coverage when workers lose employer-tied insurance.
- Designed and executed a propensity-score-matched event-study difference-in-differences on a person-month panel of 5.9 million observations from the Medical Expenditure Panel Survey (MEPS, 2004–2018), with a two-part hurdle decomposition of premium burden and latent class analysis of pre-displacement job profiles.
- Found the ACA absorbed roughly one-quarter of the two-month coverage loss and one-third of the four-month loss, with gains concentrated in a high-coverage Finance and Professional class while three precarious-work classes showed essentially no gains.

Research Experience

NYU Silver School of Social Work New York, NY
Research Assistant to Prof. Wen-Jui Han 06/2025 – 05/2026
Ford Foundation-funded project, "Work as a Social Determinant of Health: Strengthening the Case for Workers' Rights and Social Protections." Research focuses on how parental employment affects adolescent cognitive development and risk-taking behaviors.

- Developed a reproducible data-cleaning workflow in Stata for the National Longitudinal Survey of Youth 1979 (NLSY79) and its Children Supplement, cleaning approximately 11,000 records across 15 waves and constructing over 50 derived variables; implemented a multiple imputation workflow to handle missing data across waves.
- Implemented sequence cluster analysis to identify maternal employment trajectory patterns, then applied growth-curve modeling and stage-specific regression analysis to examine how nonstandard and precarious parental work schedules influence adolescent cognitive development and risk-taking behaviors.

CUNY Graduate School of Public Health New York, NY
Research Assistant to Prof. Mustafa Hussein 06/2025 – 05/2026
Research focuses on the impact of healthcare policy changes on insurance coverage and out-of-pocket premium payments.

- Built an end-to-end Stata pipeline integrating MEPS Full-Year Consolidated, Jobs, and Person-Round files (2004–2018) into a person-month panel of 301,000 individuals, constructing measures of insurance status, employment, wages, and medical expenditures.
- Conducted imputation and data wrangling across the panel to support causal-inference analyses of healthcare policy and labor outcomes.

Quanthon Corporation Taipei, Taiwan
Research Assistant (Full-time) 07/2023 – 07/2024
Think tank and data-science research consultancy providing policy analysis and business intelligence.

- Analyzed government and consumer survey data in Stata to evaluate policy initiatives and produce executive-level visualizations and reports for government officials and commercial clients.

- Designed survey instruments in Qualtrics and assisted with a methodology paper on Random Forest imputation for structural changes in longitudinal survey design.

Selected Academic Projects

Determinants of Household Debt Across OECD Economies
Group Project (Modeling Lead) · "Messy Data and Machine Learning," NYU 09/2025 – 12/2025

- Studied what predicts household debt accumulation across developed economies; applied Random Forest, LASSO, and multinomial logistic regression, with Random Forest outperforming multinomial logistic regression (91.3% vs. 76.9% accuracy) and identifying credit market accessibility and GDP as the dominant predictors, with income inequality a secondary factor.

Medicaid Expansion and Out-of-Pocket Costs: A Double Machine Learning Approach
Term Paper · "Machine Learning in Economics," NYU 02/2025 – 05/2025

- Estimated whether Medicaid expansion reduced out-of-pocket medical spending; applied a PLIV estimator in R using double machine learning (Random Forest, LASSO, XGBoost with cross-fitting) to estimate the causal effect.

Assessing the Financial and Coverage Effects of Medicaid Expansion
Group Project Lead · "Applied Statistics and Econometrics," NYU 02/2025 – 05/2025

- Tested whether Medicaid expansion improved coverage, financial protection, and health outcomes jointly; led all coding, data cleaning, and empirical strategy, estimating survey-weighted DiD and event-study models on CPS ASEC microdata and combining SUR models to test joint significance.

Additional Information

Technical Skills

Proficient	**R** (Data Wrangling, ML, API Interaction, Imputation, Visualization, Statistical & Econometric Modeling); **Stata** (Data Wrangling, Imputation, Visualization, Statistical & Econometric Modeling)
Working Knowledge	**Python** (Data Wrangling, Web-scraping); **SQL** (Data Querying); **Tableau** (Visualization); **Qualtrics** (Survey Design); **LaTeX** (Formatting)
Public Datasets	NLSY, MEPS, NSFG, IPUMS (CPS, ATUS)
Languages	English (Full Professional), Mandarin Chinese (Native)

References

Prof. Wen-Jui Han, Professor
NYU Silver School of Social Work

Prof. Mustafa Hussein, Assistant Professor
CUNY Graduate School of Public Health

Dr. Gustavo Pereira, Adjunct Professor
NYU Department of Economics

Dr. Jui-Chung Allen Li, CEO
Quanthon Corporation

References available upon request.

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